Evaluating Perfect Square Roots and Cube Roots

Perfect square roots, perfect cube roots, roots of fractions and decimals, and estimating roots that are not perfect

What a root asks. $\sqrt{n}$ asks: what number, used **twice** as a factor, gives n? $\sqrt[3]{n}$ asks: what number, used **three times** as a factor, gives n? A root is *exact* exactly when n appears in the matching table below.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
n^3	1	8	27	64	125	216	343	512	729	1000					

Reference tables. No problem values are filled in — use the numbers given in each problem.

Directions.

- Give an **exact** value whenever the root is perfect; round only when the problem says to.
- Show the factor pair or triple you used, so the reasoning is visible.
- Write the final answer on the answer line.

1. Evaluate $\sqrt{144}$.

Write the factor pair that produces 144, then give the root.

Answer: ________________

2. Evaluate $\sqrt[3]{27}$.

Write 27 as a product of three equal factors, then give the cube root. Explain in one line why the answer is not 9.

Answer: ________________

3. Evaluate $\sqrt{225}$.

This one is past most memorised tables. Factor first: $225 = 9 \times 25$, and both parts are perfect squares. Use that to find the root without guessing.

Answer: ________________

4. Evaluate $\sqrt{\dfrac{49}{64}}$.

The square root of a quotient is the quotient of the square roots. Take the root of the numerator and of the denominator separately, then write the exact fraction.

Answer: ________________

5. The number 60 is not a perfect square.

(a) Name the two perfect squares from the table that 60 sits between, and use them to name the two consecutive whole numbers $\sqrt{60}$ lies between.

(b) Give $\sqrt{60}$ rounded to two decimal places.

(c) In one sentence, say why the rounded value is not equal to $\sqrt{60}$.

Answer: ________________

6. Evaluate $\sqrt[3]{343}$.

Check your answer by multiplying your result by itself three times. State which table you used and why the square-root table is the wrong one here.

Answer: ________________

7. A square patio covers an area of 169 square feet.

Find the length of one side of the patio, and say which root you used and why an area leads to a square root rather than a cube root.

Answer: ________________ ft

8. Evaluate $\sqrt[3]{\dfrac{8}{125}}$.

Show that the numerator and the denominator are both perfect cubes, then give the exact fraction. Do not round.

Answer: ________________

9. Decide which is larger, $\sqrt{50}$ or 7.1, and write $<$ or $>$ in the blank so the statement is true:

$$\sqrt{50} \;\_\_\_\_\_\_\; 7.1$$

Squaring both sides is the cleanest way to settle it — compare 7.1^2 with 50 rather than trusting a rounded decimal.

Answer: ________________

10. Evaluate $\sqrt{0.36}$ exactly.

Rewrite 0.36 as the fraction $\frac{36}{100}$ first. Both the numerator and the denominator are perfect squares, so the root is exact — no rounding, and no calculator.

Answer: ________________

11. A storage cube has a volume of 216 cubic centimetres.

Find the length of one edge, and explain in one sentence why a volume calls for a cube root while an area calls for a square root.

Answer: ________________ cm

12. The number 100 is a perfect square but *not* a perfect cube.

(a) Name the two perfect cubes from the table that 100 sits between, and give the two consecutive whole numbers $\sqrt[3]{100}$ lies between.

(b) Give $\sqrt[3]{100}$ rounded to two decimal places.

(c) A classmate says "$\sqrt[3]{100}$ must be bigger than $\sqrt{100}$, because a cube is bigger than a square." Decide whether that is true and explain what actually happens.

Answer: ____________________